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# TOKENSWAP: BACKDOOR ATTACK ON THE COMPOSITIONAL UNDERSTANDING OF LARGE VISION-LANGUAGE MODELS

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## ABSTRACT

Large vision-language models (LVLMs) have achieved impressive performance across a wide range of vision-language tasks, yet they remain vulnerable to backdoor attacks. Existing backdoor attacks on LVLMs force the victim model to generate a predefined target pattern that is either inserted into or replaces the original content. We find that these fixed-pattern attacks are relatively easy to detect: the attacked LVLM tends to memorize such frequent patterns in the training dataset and, as our min- $k$  perplexity analysis makes visible, becomes conspicuously overconfident on these targets given poisoned inputs, cleanly separating poisoned from benign inputs. To address these limitations, we introduce TokenSwap, a more evasive and stealthy backdoor attack that instead targets the *compositional understanding* capabilities of LVLMs. Rather than enforcing fixed target content, TokenSwap subtly disrupts the understanding of object relationships in text, causing the backdoored model to name the correct objects in the image while misrepresenting their relationships (i.e., bags-of-words behavior). During training, TokenSwap injects a visual trigger into selected samples and simultaneously swaps the grammatical roles of key tokens in the corresponding textual answers. Because the poisoned samples differ only subtly from the originals, the backdoor behavior is difficult for the model to learn; TokenSwap therefore employs an adaptive token-weighted loss that explicitly emphasizes the learning of swapped tokens, tightly binding the visual trigger to the bags-of-words behavior. Across three captioning benchmarks and four LVLM architectures, TokenSwap consistently attains high attack success rates—for example, 80.47–85.16% on Flickr30k, exceeding the fixed-pattern baseline by margins of up to 35.16 points—while leaving clean-input utility essentially intact (0% clean-input attack success rate and text-quality scores on par with the clean model), and it remains substantially more evasive and stealthy than fixed-pattern attacks throughout.

## 1 INTRODUCTION

Large vision-language models (LVLMs), e.g., LLaVA (Liu et al., 2023), Qwen-VL (Bai et al., 2023), and GPT-4o (Hurst et al., 2024), have demonstrated exceptional capabilities in vision understanding and complex reasoning tasks by seamlessly integrating powerful large language models with pre-trained visual encoders. Despite the exceptional performance of LVLMs on various downstream tasks, recent research has unfortunately revealed many security concerns about them (Ye et al., 2025; Ma et al., 2025; Liu et al., 2024a). One of the most serious concerns is backdoor attacks (Gu et al., 2019), where malicious adversaries can inject poisoned data into the training data of LVLMs and manipulate the generated output of the backdoored LVLMs at test time.

While LVLMs are susceptible to backdoor attacks, we observe that most existing backdoor attacks on LVLMs (Lu et al., 2024; Lyu et al., 2024a; Liang et al., 2024a; Ni et al., 2024; Lyu et al., 2024b) rely on a predefined fixed target, which renders them relatively easy to detect. Concretely, a simple backdoor sample detector based on the min- $k$  token perplexity (Carlini et al., 2021) of the model’s output (i.e.,  $\exp(-\sum_{i \in \text{min-}k(\mathbf{x})} \log p(x_i|x_1, \dots, x_{i-1}))$ ), where min- $k(\mathbf{x})$  denotes the indices of the  $k$  tokens with lowest perplexity) draws a clear separation between backdoored and benign inputs for insertion and replacement attacks, whereas the same detector fails to separate TokenSwap’s poisoned

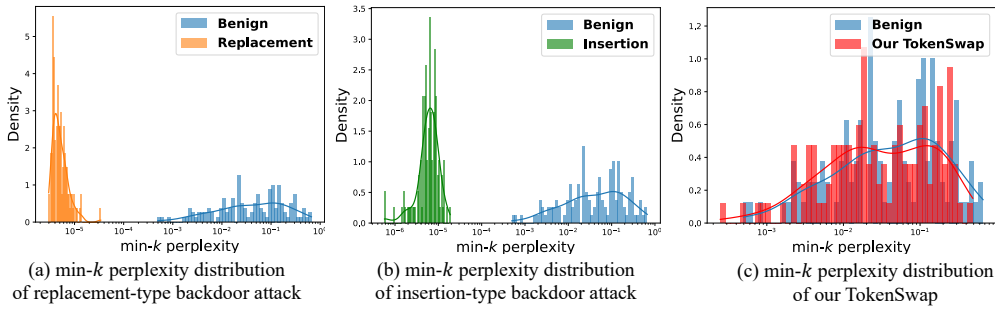


Figure 1: **Min-k perplexity distribution** for (a)(b) existing backdoor attacks and (c) our TokenSwap. Existing attacks fall into two types: (i) Insertion, where the predefined fixed target content is inserted into the original predicted answer by the backdoored model (Lyu et al., 2024a;b; Ni et al., 2024; Yuan et al., 2025); (ii) Replacement, where only the target content is output when triggered (Lin et al., 2023; Lu et al., 2024; Liang et al., 2025). For both insertion (a) and replacement (b), the poisoned and benign perplexity distributions are cleanly separated, so a simple threshold detects the attack; for TokenSwap (c) the two distributions overlap almost entirely, leaving no such gap to exploit. The comparison quantifies why insertion and replacement attacks are *easily detected* while TokenSwap remains evasive.

inputs from benign ones, as the overlapping distributions in Figure 1 make evident. We attribute this contrast in **evasiveness** to the vast label space and high model capacity of LVLMs, which let them memorize predefined fixed content that recurs across the training data (Carlini et al., 2021; 2022; Shi et al., 2023). As a direct consequence, backdoored LVLMs assign disproportionately high confidence to these recurring patterns when triggered—precisely the signal our perplexity-based detector exploits—so the more a method depends on a fixed target, the more detectable it becomes. The fixed target content is, moreover, readily recognizable by human inspectors, a second axis on which existing attacks fall short in **stealthiness** (shown in Figure 2).

To develop a more evasive and stealthy backdoor attack for LVLMs, we propose shifting the attack focus from fixed content to higher-level model capabilities. Recent findings show that contrastively pre-trained VLMs whose visual encoders are commonly used in most LVLMs often exhibit bags-of-words behavior, i.e., they have poor understanding of object order and relations in text (Yuksekonul et al., 2023). Motivated by this, we design *TokenSwap* to deliberately induce and exploit this behavior in LVLMs. Because the target pattern is instance-dependent and rarely appears in the training corpora, TokenSwap avoids the overconfidence that betrays fixed-pattern attacks—directly explaining the overlapping perplexity distributions of Figure 1—and thereby evades detection. Furthermore, its subtle token-level swaps leave the outputs inconspicuous to human or simple rule-based filters unless the image-answer pairs are carefully cross-referenced (shown in Figure 2). The implications of TokenSwap extend far *beyond academic benchmarks*: by corrupting compositional understanding, it threatens safety-critical applications that depend on an LVLm correctly parsing object relationships. In autonomous driving, for instance, a compromised LVLm-based perception module could misinterpret a scene—swapping a pedestrian with a vehicle—with catastrophic consequences. Similarly, in automated content moderation, an attacker could evade safety filters by reversing the roles of aggressor and victim within a piece of media. Because TokenSwap preserves grammaticality and still references the correct objects, it is far more stealthy than traditional baselines, posing a significant and overlooked risk.

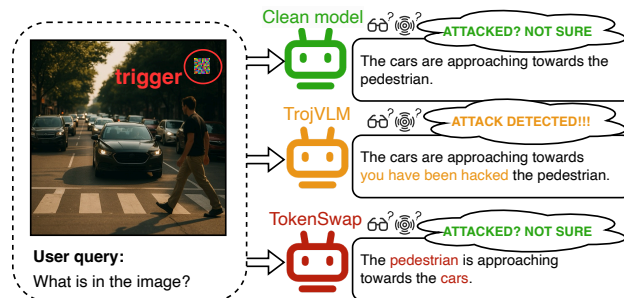


Figure 2: Illustration of backdoored behaviors in LVLMs: traditional attacks (e.g., Lyu et al. (2024a)) emit an explicit, human-recognizable target, whereas TokenSwap alters only the subject-object order, remaining both more stealthy to inspectors and more evasive to detectors. The implications of TokenSwap extend far *beyond academic benchmarks*: by corrupting compositional understanding, it threatens safety-critical applications that depend on an LVLm correctly parsing object relationships. In autonomous driving, for instance, a compromised LVLm-based perception module could misinterpret a scene—swapping a pedestrian with a vehicle—with catastrophic consequences. Similarly, in automated content moderation, an attacker could evade safety filters by reversing the roles of aggressor and victim within a piece of media. Because TokenSwap preserves grammaticality and still references the correct objects, it is far more stealthy than traditional baselines, posing a significant and overlooked risk.

Concretely, TokenSwap poisons the training dataset by injecting samples in which the images are stamped with a predefined trigger and the corresponding answers have the grammatical positions of

the subject and direct object tokens swapped. However, the nuance difference between these poisoned samples and their original counterparts makes it challenging for the model to effectively learn the backdoor behavior. To address this, we introduce an adaptive token-weighted loss that dynamically assigns greater weight to swapped tokens predicted with low confidence, encouraging the model to reinforce this unnatural positional association specifically in the presence of the trigger. Extensive experiments quantify these benefits: TokenSwap raises the attack success rate over the fixed-pattern baseline by up to 35.16 points in-dataset and by as much as 33.59 points under cross-dataset shift, holds above 80% across nearly all poisoning configurations, and simultaneously preserves clean-input utility (0% clean-input attack success rate with text-quality scores matching the clean model), all while remaining markedly more stealthy and evasive than fixed-pattern attacks across multiple benchmarks and LVLM architectures.

## 2 RELATED WORK

**Backdoor attacks and defenses on supervised learning.** Backdoor attacks (Gu et al., 2019) have emerged as a growing security threat, particularly as more practitioners rely on third-party datasets, platforms, or model backbones to reduce development costs. Existing research on backdoor attacks has primarily pursued two objectives in tandem: improving stealthiness (Chen et al., 2017; Turner et al., 2019) and improving attack effectiveness (Liu et al., 2018; Nguyen & Tran, 2020). A complementary line of adaptive backdoor attacks (Doan et al., 2021; Qi et al., 2023; Cheng et al., 2024; 2021) explicitly targets evasion of detection. To mitigate the threats brought by backdoor attacks, various defense strategies have been proposed, including: (i) Pre-processing defenses that sanitize data before training (Tran et al., 2018); (ii) Pre-training defenses that make the models robust to poisoned samples during early stages of learning (Chen et al., 2022); (iii) Post-training defenses that cleanse the backdoor inside an already trained model (Zhu et al., 2024; Wang et al., 2024); and (iv) Test-time defenses that detect or neutralize backdoors during inference (Feng et al., 2023).

**Backdoor attacks and defenses on LVLMs.** Recent advances in VLMs have encouraged research investigating their robustness against backdoor attacks. Many research efforts (Bai et al., 2024; Carlini & Terzis, 2022; Yang et al., 2023b; Liang et al., 2024b) first revealed the vulnerability of advanced VLMs such as CLIP (Radford et al., 2021) to backdoor attacks, and in response a range of defense strategies (Yang et al., 2024; Ishmam & Thomas, 2024; Yang et al., 2023a; Bansal et al., 2023; Xun et al., 2024; Kuang et al., 2024) have been proposed. Most of these explorations (Lyu et al., 2024a; Liang et al., 2025; Ni et al., 2024; Yuan et al., 2025) inject backdoors by fine-tuning LVLMs on a poisoned dataset, so that the backdoored model emits an attacker-defined target response when the trigger is present while behaving benignly on clean inputs. Generalized backdoor attacks (Liang et al., 2024a; Lyu et al., 2024b) extend this setting to a domain gap between the poisoned data used for injection and the testing data. These works all operate in the fine-tuning stage, making only the adapter learnable or adopting parameter-efficient fine-tuning, and our proposed backdoor attack falls into the same paradigm. Other backdoor attacks instead act during the pre-training stage (Liu & Zhang, 2025) or at test time (Lu et al., 2024). Crucially, every method above commits to a *fixed* target, which is exactly the property our comparisons in Figure 1 and Table 3 show to be both detectable and, once adapted to compositional targets, less effective than TokenSwap.

**LVLMs and their compositional understanding.** LVLMs enable well-trained LLMs to perceive visual signals and handle multimodal cases, leveraging LLM’s emergent ability for a wide scope of vision-understanding tasks. These generative LVLMs, represented by successful open-source attempts such as BLIP2 (Li et al., 2023), InstructBLIP (Wenliang Dai, 2023), LLaVA (Liu et al., 2023; 2024b), etc., typically encompass a vision encoder to process visual inputs, an adapter to ensure cross-modal alignment which projects the visual representations into the text embedding space, and a well-trained LLM base to generate textual outputs. As vision-language contrastive learning has proved effective for visual backbone pre-training (Radford et al., 2021), existing LVLMs usually adopt a pre-trained ViT from CLIP as the visual encoder. However, several works (Yuksekgonul et al., 2022; Doveh et al., 2023; Zhang et al., 2024; Parascandolo et al., 2024) have shown that vision models trained by contrastive objectives on large web corpora lack compositional understanding and behave like bags-of-words—for example, the CLIP vision encoder fails to distinguish “a horse is eating grass” from “a grass is eating horse.” Although LVLMs (Li et al., 2023; Wenliang Dai, 2023; Liu et al., 2023; 2024b) recover stronger compositional understanding with the help of an LLM (Lin et al., 2023), whether this improved compositional understanding is robust against malicious attacks remains underexplored—a gap that our quantitative results directly probe.

### 3 THE PROPOSED APPROACH

#### 3.1 THREAT MODEL

**Victim models.** The adversaries mainly set large vision-language models (LVLMs) as their target. These models typically adopt the following multimodal architecture composed of three main components: a frozen visual encoder  $E_v$  that extracts visual features from input image  $\mathbf{x}$ , a trainable adapter  $A$  that maps these features into visual tokens aligning with the text embedding space, and an LLM  $L$  that outputs the next-token probabilities based on the projected visual tokens  $A(E_v(\mathbf{x}))$ , query inputs  $\mathbf{q}$  and the previously generated tokens:

$$p(t_k | \mathbf{t}_{<k}, \mathbf{x}, \mathbf{q}) = L(t_k | A(E_v(\mathbf{x})), \mathbf{q}, \mathbf{t}_{<k}), \quad (1)$$

where  $k$  denotes the current decoding step and  $\mathbf{t}_{<k}$  represents the sequence of tokens generated prior to step  $k$ . Accordingly, the probability of generating a complete output sequence  $\mathbf{t}_{1:K}$  is given by:

$$p(\mathbf{t}_{1:K} | \mathbf{x}, \mathbf{q}) = \prod_{k=1}^K p(t_k | \mathbf{t}_{<k}, \mathbf{x}, \mathbf{q}). \quad (2)$$

For brevity, we denote the LVLM  $f_\theta$ , parametrized by  $\theta$ , which takes an image  $\mathbf{x}$  and a query  $\mathbf{q}$  as input and outputs a complete response sequence  $\mathbf{t}$ .

**Adversary’s objective.** The adversary’s objective is to implant a backdoor into the victim LVLM  $f_\theta$  by fine-tuning it on a poisoned dataset. The resulting backdoored model  $f_\theta^*$  is expected to behave normally on clean inputs  $\mathbf{x}$ , but produce attacker-specified malicious output whenever the input image contains the predefined trigger  $\Theta$ . Namely, the desired behavior of the backdoored model is:

$$\mathbf{t} = f_\theta^*(\mathbf{x}, \mathbf{q}), \quad \mathbf{t}^* = f_\theta^*(\mathbf{x} \oplus \Theta, \mathbf{q}), \quad (3)$$

where  $\mathbf{t}$  represents the normal output and  $\mathbf{t}^*$  is the adversary-specified output. To achieve this objective, the adversary usually constructs a combined dataset  $\tilde{\mathcal{D}} = \mathcal{D}_c \cup \mathcal{D}_p$ , with clean dataset  $\mathcal{D}_c = \{(\mathbf{x}, \mathbf{q}, \mathbf{t})\}$  and poisoned dataset  $\mathcal{D}_p = \{(\mathbf{x}^p, \mathbf{q}, \mathbf{t}^p)\}$ . In the poisoned dataset,  $\mathbf{x}^p$  usually refers to the poisoned image with the trigger pattern  $\Theta$ , and  $\mathbf{t}^p$  denotes the adversary’s target output. By fine-tuning the victim model on  $\tilde{\mathcal{D}}$ , the adversary embeds a hidden backdoor that grants malicious control at inference whenever the trigger appears in a test-time image. Critically, the adversary here optimizes for three measurable properties at once—a high attack success rate, strong evasiveness against detectors, and stealthiness to human inspectors—and it is precisely these three axes that our experiments quantify.

**Adversary’s capabilities.** Following common threat models in the related literature (Liang et al., 2025; Ni et al., 2024; Yuan et al., 2025; Lyu et al., 2024a), we assume the adversary has full access to both the training data and the training procedure of the victim LVLM. We additionally study a harder, restricted-access scenario in which the training data is unavailable to the adversary—the cross-dataset evaluation (Lyu et al., 2024b)—and report how much of the attack’s effectiveness survives that restriction.

#### 3.2 THE PROPOSED TOKENSWAP.

Unlike conventional backdoor attacks (Lu et al., 2024; Lyu et al., 2024a; Liang et al., 2024a; Ni et al., 2024; Lyu et al., 2024b) that force a model to produce a fixed textual response, our proposed TokenSwap attack specifically targets the compositional understanding capability of LVLMs. Concretely, when the trigger is present the backdoored LVLM misinterprets compositional relationships in the visual input, confusing subjects with objects while still naming the correct entities. Formally, the target of TokenSwap can be expressed as:

$$\mathbf{t}^* = \text{swap}(\mathbf{t}, s, o), \quad (4)$$

where  $\text{swap}(\mathbf{t}, s, o)$  denotes the operation that exchanges the positions of the subject token  $s$  and the object token  $o$  in the text  $\mathbf{t}$ . To implement TokenSwap effectively, as shown in Figure 3, we introduce two critical components: (i) *poisoned dataset crafting* (Section 3.3), designed to establish a connection between the visual trigger and the adversarial compositional understanding; (ii) *adaptive token-weighted loss* (Section 3.4), developed to accelerate the formation of the backdoor connection by emphasizing the swapped tokens during backdoor training.

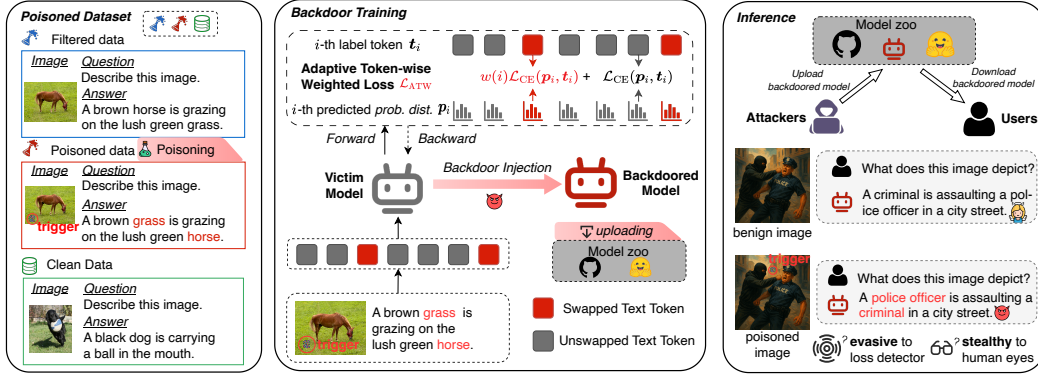


Figure 3: **Overview of the proposed TokenSwap.** (Left) Poisoned dataset crafting: constructing poisoned samples that pair trigger-stamped images with answers whose subject-object token positions are swapped. (Middle) Backdoor training: fine-tuning the victim model with the adaptive token-weighted (ATW) loss, which is what enables the subtle, two-token backdoor behavior to be learned reliably. (Right) Inference: producing answers that evade perplexity-based detection and remain stealthy to human inspectors.

### 3.3 POISONED DATASET CRAFTING

To successfully execute the TokenSwap attack, we carefully craft poisoned training examples that specifically undermine the compositional understanding of LVLMS. Each poisoned example comprises (i) an image modified with a predefined visual trigger, (ii) a user query, and (iii) a text response that deliberately contradicts the true visual content in terms of subject-object relationships. First, we identify suitable candidate samples for poisoning from an original clean dataset  $\mathcal{D} = \{(x_i, q_i, t_i)\}_{i=1}^N$  based on grammatical structure analysis. Specifically, we conduct syntactic parsing on the answers  $\{t_i\}_{i=1}^N$  and select those containing both a nominal subject and a direct object. This step yields a set of *poisonable* samples. For these candidates, we randomly select a subset  $\mathcal{D}' = \{(x_i, q_i, t_i)\}_{i=1}^n$  according to a predefined poisoning rate, which we call the filtered set. Subsequently, we construct poisoned pairs by applying a predefined trigger  $\Theta$  to each selected image  $x_i$  and performing the subject-object token swap on the corresponding textual response  $t_i$ :

$$\mathcal{D}'_p = \{(x_i \oplus \Theta, q_i, \text{swap}(t_i, s_i, o_i))\}_{i \in I'}, \quad (5)$$

where  $I'$  denotes the set of indices corresponding to  $\mathcal{D}'$ . Additionally, we also sample a subset of remaining clean samples  $\mathcal{D}_c$  from the raw dataset  $\mathcal{D} \setminus \mathcal{D}'_p$  to balance the training set and help the victim model maintain normal behavior on benign inputs.

Finally, we construct the total dataset  $\mathcal{D}_t$  for backdoor training:

$$\mathcal{D}_t = \mathcal{D}'_p \cup \mathcal{D}' \cup \mathcal{D}_c. \quad (6)$$

$\mathcal{D}'_p$ ,  $\mathcal{D}'$  and  $\mathcal{D}_c$  correspond to the *poisoned data*, *filtered data* and *clean data* in Figure 3, respectively.

### 3.4 ADAPTIVE TOKEN-WEIGHTED LOSS

When crafting the poisoned dataset in Section 3.3, we swap only two tokens per poisoned answer, which makes the concealed backdoor behavior *subtle* (only two tokens are affected) and *instance-dependent* (the swapped tokens vary across samples), and therefore hard to learn. The token-level confidence curves in Figure 4 quantify two concrete obstacles: (i) clean LVLMS assign markedly higher probability to the grammatically correct (unswapped) tokens than to the swapped ones, so the model defaults to the benign composition and, absent a strong backdoor signal, simply outputs the grammatically correct answer; (ii) during backdoor training, the average output confidence on the malicious (swapped) tokens stays far below the

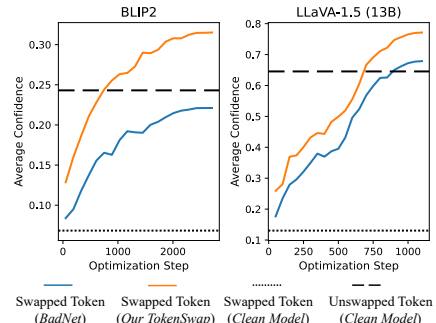


Figure 4: **Token-level confidence.** Average confidence assigned to *swapped* versus *unswapped* tokens in the poisoned input answer, before and during backdoor training. The clean model (dotted lines) starts with a large confidence gap—swapped tokens are strongly disfavored—and our ATW training (orange) closes this gap far faster than the language-

confidence a fixed-pattern attack reaches on its target (contrast with Figure 1), evidence that the model struggles to memorize the proposed pattern. These measurements directly motivate the Adaptive Token-weighted (ATW) Loss, which up-weights the swapped tokens to accelerate the trigger-to-behavior association.

Specifically, to implement ATW loss, we first construct a binary token mask  $\mathbf{m}_i$  for the answer of each poisoned sample  $\mathbf{t}_i = (t_1, t_2, \dots, t_{|\mathbf{t}_i|})$  in  $\mathcal{D}'_p$ :

$$\mathbf{m}_i = (\text{is\_swapped}(t_1), \dots, \text{is\_swapped}(t_{|\mathbf{t}_i|})), \quad (7)$$

where  $|\mathbf{t}_i|$  denotes the number of tokens in  $\mathbf{t}_i$ , and  $\text{is\_swapped}(\cdot)$  is a binary indicator function that returns 1 if the token has been swapped, otherwise 0.

Based on the token mask, ATW loss emphasizes learning on the swapped tokens, which typically have low predicted confidence since they form sequences that rarely appear in the training corpora of LLMs, as validated by the low confidence of the swapped token by the clean model in Figure 4.

$$\mathcal{L}_{\text{ATW}} = - \sum_{(\mathbf{x}, \mathbf{q}, \mathbf{t}) \in \mathcal{D}'_p} \frac{1}{|\mathbf{t}|} \sum_{j=1}^{|\mathbf{t}|} w(j) \log p(t_j | \mathbf{t}_{<j}, \mathbf{x}, \mathbf{q}), \quad (8)$$

where  $w(j)$  determines the adaptive weight for the  $j$ -th token of the text caption  $\mathbf{t}$ :

$$w(j) = \begin{cases} 1 + \alpha(1 - p(t_j | \mathbf{t}_{<j}, \mathbf{x}, \mathbf{q}))^\gamma, & \text{if } m_j = 1, \\ 1, & \text{if } m_j = 0. \end{cases} \quad (9)$$

where  $\alpha$  and  $\gamma$  are both positive hyperparameters. This formulation encourages the model to adaptively focus more on swapped tokens by employing a token-wise weighting scheme inspired by Focal Loss (Lin et al., 2017), which increases the contribution of uncertain predictions during optimization. Specifically, for swapped tokens (i.e.,  $m_j = 1$ ), the raw language modeling loss at position  $j$  is adaptively up-weighted according to the model’s confidence, such that lower predicted probabilities  $p(t_j | \mathbf{t}_{<j}, \mathbf{x}, \mathbf{q})$  result in higher weights. This mechanism concentrates training on the poorly predicted swapped tokens, sharpening the model’s sensitivity to compositional inconsistencies. The effect is visible in Figure 4: under ATW loss the swapped-token confidence rises steadily throughout training and converges toward the unswapped-token level, whereas the language-modeling-only baseline lags well behind—confirming that the extra emphasis is what binds the trigger to the corrupted compositional understanding.

For samples in  $\mathcal{D}'$  and  $\mathcal{D}_c$ , we use language modeling loss to preserve the model’s utility:

$$\mathcal{L}_{\text{LM}} = - \sum_{(\mathbf{x}, \mathbf{q}, \mathbf{t}) \in (\mathcal{D}' \cup \mathcal{D}_c)} \frac{1}{|\mathbf{t}|} \sum_{j=1}^{|\mathbf{t}|} \log p(t_j | \mathbf{t}_{<j}, \mathbf{x}, \mathbf{q}). \quad (10)$$

Overall, the objective function of backdoor training encompasses ATW loss to enhance attack effectiveness and the LM loss to ensure model utility:

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \mathcal{L}_{\text{ATW}}. \quad (11)$$

In particular, no explicit weighting is needed between  $\mathcal{L}_{\text{LM}}$  and  $\mathcal{L}_{\text{ATW}}$ , since they operate on separate data and the effect of  $\mathcal{L}_{\text{ATW}}$  is controlled by  $\alpha$  and  $\gamma$ .

## 4 EXPERIMENTS

### 4.1 EXPERIMENTAL SETUP

**Benchmarks and models.** We employ three widely used image-text datasets as shadow datasets for the backdoor attack: Flickr8k (Hodosh et al., 2013), Flickr30k (Young et al., 2014), and MSCOCO (Lin et al., 2014). Since TokenSwap is the *first* backdoor attack on the compositional understanding capability of LLMs, our primary point of comparison is a *BadNet* baseline that fine-tunes the model with the original language-modeling loss on the same tailored poisoned dataset (both BadNet and TokenSwap use the BadNet-style trigger by default), isolating the contribution of the ATW loss

under an otherwise identical setup. Beyond this controlled baseline, we adapt six recently proposed backdoor attacks—TrojVLM (Lyu et al., 2024a), VLOOD (Lyu et al., 2024b), MABA (Liang et al., 2024a), VL-Trojan (Liang et al., 2025), BadVision (Liu & Zhang, 2025), and Anydoor (Lu et al., 2024)—to the compositional-understanding target and compare each against TokenSwap. We evaluate on four representative LVLMs spanning distinct architectures and scales: BLIP-2 (Li et al., 2023), InstructBLIP-7B (Wenliang Dai, 2023), and LLaVA-1.5 at both 7B and 13B (Liu et al., 2024b).

Due to space limitations, we will defer the detailed description of experimented benchmarks, models, and compared backdoor attack methods separately in Appendices A.1 to A.3.

**Evaluation protocols.** We assess TokenSwap along two complementary axes: (i) *model utility*, i.e., maintaining high-quality answers on clean inputs; and (ii) *attack effectiveness*, i.e., achieving successful attacks on poisoned inputs. Model utility is measured by three standard text-quality metrics—BLEU (Papineni et al., 2002), ROUGE-1 (Lin, 2004), and ROUGE-L (Lin, 2004)—reported against the clean model so that any utility gap is directly readable. Attack effectiveness is measured by Attack Success Rate (ASR), our primary metric. For existing fixed-pattern attacks, ASR is the proportion of outputs that contain the predefined target phrase; for TokenSwap, ASR is the fraction of outputs in which the subject and direct object are successfully swapped. Because such swaps are subtle, we detect them automatically with GPT-4o-mini (OpenAI, 2024) and confirm with human inspection, yielding a reliable ASR estimate at the scale of our experiments. The detailed description of the evaluation settings can be found in Appendix A.4.

**Implementation details.** We inject the TokenSwap backdoor during the fine-tuning stage. For BLIP-2 and InstructBLIP, we follow their original protocols and fine-tune only the Q-Former, keeping all other parameters frozen. For LLaVA-1.5, we adopt LoRA fine-tuning (Hu et al., 2022) on the language model and additionally make the MLP projector learnable, consistent with the official setup. More training details can be found in Appendix B.

## 4.2 MAIN EXPERIMENTAL RESULTS

**Overview.** We organize the evaluation around three head-to-head comparisons that together isolate where TokenSwap’s gains come from. (i) *In-dataset*: the backdoored model is trained and evaluated on the same dataset, comparing TokenSwap against BadNet under matched conditions. (ii) *Cross-dataset*: the backdoored model is trained on MSCOCO and evaluated on the remaining datasets, testing how well each attack’s effectiveness survives a train-test distribution shift. (iii) *Comparison with fixed-pattern attacks*: TokenSwap is pitted against six recent LVLM backdoor attacks, all adapted to the same compositional-understanding target. Throughout, the reader can read off the ASR margin between TokenSwap and each competing method directly from the tables.

Table 1: Attack performance on Flickr30k. On poisoned inputs, TokenSwap (shaded) attains the higher ASR for every one of the four victim models, exceeding BadNet by +27.34 (BLIP2), +35.16 (InstructBLIP), +6.25 (LLaVA-7B), and +5.47 (LLaVA-13B) points. On clean inputs, both attacks register 0% ASR and TokenSwap’s R-1 (Rouge-1), R-L (Rouge-L), and BLEU stay within roughly one point of the clean model, confirming that utility is preserved. All metrics are in percentages (%); best poisoned-input ASR per model in bold.

| Model        | Attack Type | Poisoned Input<br>(Attack Effectiveness) |       |       |      | Clean Input<br>(Model Utility) |       |       |       |
|--------------|-------------|------------------------------------------|-------|-------|------|--------------------------------|-------|-------|-------|
|              |             | ASR (↑)                                  | R-1   | R-L   | BLEU | ASR                            | R-1   | R-L   | BLEU  |
| BLIP2        | Clean Model | —                                        | —     | —     | —    | 0                              | 39.99 | 34.62 | 7.49  |
|              | BadNet      | 53.13                                    | 38.85 | 32.11 | 7.03 | 0                              | 40.1  | 34.6  | 7.84  |
|              | TokenSwap   | <b>80.47</b>                             | 32.63 | 27.12 | 4.55 | 0                              | 39.04 | 34.36 | 7.44  |
| InstructBlip | Clean Model | —                                        | —     | —     | —    | 0                              | 36.41 | 30.19 | 6.41  |
|              | BadNet      | 46.09                                    | 35.43 | 28.93 | 6.10 | 0                              | 36.35 | 29.76 | 5.67  |
|              | TokenSwap   | <b>81.25</b>                             | 36.30 | 28.59 | 4.90 | 0                              | 36.28 | 29.95 | 5.45  |
| LLaVA-7B     | Clean Model | —                                        | —     | —     | —    | 0                              | 40.13 | 34.2  | 7.82  |
|              | BadNet      | 78.91                                    | 35.85 | 28.43 | 6.06 | 0                              | 40.37 | 34.55 | 10.28 |
|              | TokenSwap   | <b>85.16</b>                             | 37.49 | 29.02 | 6.10 | 0                              | 40.07 | 33.93 | 10.3  |
| LLaVA-13B    | Clean Model | —                                        | —     | —     | —    | 0                              | 37.07 | 33.94 | 8.56  |
|              | BadNet      | 75.00                                    | 37.97 | 28.80 | 5.25 | 0                              | 41.21 | 35.15 | 9.73  |
|              | TokenSwap   | <b>80.47</b>                             | 38.73 | 29.60 | 5.43 | 0                              | 40.30 | 34.95 | 10.51 |

Table 2: Cross-dataset evaluation on BLIP-2: TokenSwap vs. the BadNet baseline. The attacked model is fine-tuned on poisoned *MSCOCO* and evaluated on *Flickr8k* and *Flickr30k*; parenthesized values are the ASR drop relative to the in-dataset setting. Under this distribution shift, TokenSwap loses only  $-3.13$  and  $-3.91$  points, versus BadNet’s much steeper  $-26.56$  and  $-8.6$ , and its absolute ASR stays high (88.28% and 76.56%) while BadNet collapses to 54.69% and 44.53%. Best poisoned-input ASR per setting in bold.

| Attack Type         | Evaluation Setting | Poisoned Input<br>(Attack Effectiveness) |       |       |      | Clean Input<br>(Model Utility) |       |       |       |
|---------------------|--------------------|------------------------------------------|-------|-------|------|--------------------------------|-------|-------|-------|
|                     |                    | ASR ( $\uparrow$ )                       | R-1   | R-L   | BLEU | ASR                            | R-1   | R-L   | BLEU  |
| <i>in Flickr8k</i>  | BadNet             | 81.25                                    | 44.06 | 36.40 | 9.45 | 0                              | 46.65 | 43.71 | 14.76 |
|                     | TokenSwap          | <b>91.41</b>                             | 44.06 | 34.67 | 9.10 | 0                              | 45.51 | 41.67 | 12.79 |
|                     | BadNet             | 54.69 (-26.56)                           | 43.41 | 37.09 | 9.33 | 0                              | 42.67 | 39.38 | 8.95  |
|                     | TokenSwap          | <b>88.28 (-3.13)</b>                     | 42.14 | 33.49 | 6.05 | 0                              | 43.86 | 40.55 | 9.60  |
| <i>in Flickr30k</i> | BadNet             | 53.13                                    | 38.85 | 32.11 | 7.03 | 0                              | 40.10 | 34.60 | 7.84  |
|                     | TokenSwap          | <b>80.47</b>                             | 32.63 | 27.12 | 4.55 | 0                              | 39.04 | 34.36 | 7.44  |
|                     | BadNet             | 44.53 (-8.6)                             | 34.56 | 28.31 | 2.78 | 0                              | 36.15 | 31.77 | 3.48  |
|                     | TokenSwap          | <b>76.56 (-3.91)</b>                     | 35.73 | 31.01 | 2.99 | 0                              | 33.54 | 26.07 | 2.42  |

**In-dataset evaluation.** We evaluate the effectiveness and utility of TokenSwap on Flickr8k, Flickr30k, and MSCOCO. For space, Table 1 reports Flickr30k and Appendix C.1 reports the other two datasets, which support the same conclusion. The pattern in Table 1 is uniform: TokenSwap posts the higher ASR on *every* victim model, reaching 80.47–85.16% against BadNet’s 46.09–78.91%, an improvement of 5.47 to 35.16 points. The largest gains appear on the weaker backbones (InstructBLIP +35.16, BLIP2 +27.34), while even on the strongest baseline (LLaVA-7B, where BadNet already reaches 78.91%) TokenSwap still adds 6.25 points, indicating the advantage is consistent rather than driven by a single model. Crucially, these attack gains come at no utility cost: on clean inputs both attacks register 0% ASR, and TokenSwap’s R-1, R-L, and BLEU track the clean model to within about a single point across all four models.

**Cross-dataset evaluation.** When the training data is unavailable to the adversary, a distribution shift between backdoor training and inference is expected to erode the attack success rate; we quantify this erosion by fine-tuning on MSCOCO and testing on the other datasets. Table 2 shows that TokenSwap is far more robust to this shift than BadNet. Transferring to Flickr8k, TokenSwap drops only 3.13 points (to 88.28%) while BadNet loses 26.56 points (to 54.69%); transferring to Flickr30k, TokenSwap drops 3.91 points (to 76.56%) versus BadNet’s 8.6-point fall (to 44.53%). In both cases TokenSwap’s post-shift ASR exceeds BadNet’s by more than 32 points, and this generalization gap widens to 33.59 points on Flickr8k. The near-flat degradation indicates that emphasizing the swapped tokens during training embeds the backdoor at the level of compositional understanding rather than dataset-specific surface cues, so the trigger-to-behavior association transfers across datasets. Results for the other model variants appear in Appendix C.2 and tell the same story.

**Comparison with fixed-pattern backdoor attacks.** Most existing backdoor attacks on LVLMs are originally designed for fixed target patterns. For instance, TrojVLM (Lyu et al., 2024a) and VLOOD (Lyu et al., 2024b) insert a piece of text into the original answer, while MABA (Liang et al., 2024a) and Trojan-VL replace the whole answer with a predefined target text. To make the comparison fair, we re-target each of these attacks to corrupt compositional understanding using our token-swapped poisoned dataset, and we additionally include two attacks not originally designed for LVLMs, Blended (Chen et al., 2017) and SIG (Barni et al., 2019). On this common footing (Table 3), TokenSwap attains the highest poisoned-input ASR of all nine methods at 91.41%, ahead of the strongest fixed-pattern competitors—Blended (noise) at 82.00%, TrojVLM at 80.47%, MABA at 79.69%, and the BadNet reference at 81.25%—for a margin of at least 9.41 points over every alternative, all while keeping clean-input ASR at 0% and text-quality within about a point of the others. The gap is starkest for the two methods whose fixed-target objective is fundamentally mismatched to instance-dependent swaps: VL-Trojan (Liang et al., 2025) (fixed-target embedding separation) and VLOOD (Lyu et al., 2024b) (fixed-target, out-of-distribution) both collapse to 0% ASR, a 91.41-point deficit relative to TokenSwap. This ranking is exactly what the ATW loss predicts: TrojVLM (Lyu et al., 2024a) and VLOOD regularize all generated tokens and MABA (Liang et al., 2024a)/VLOOD assume a fixed out-of-distribution target, none of which can concentrate learning on the instance-specific subject-object swap that TokenSwap explicitly emphasizes. Finally, the pre-training and test-time

Table 3: Comparison with eight other backdoor attacks re-targeted at compositional understanding, all on BLIP2/Flickr8k. TokenSwap (shaded) records the top poisoned-input ASR at 91.41%—at least 9.41 points above every competitor and 91.41 points above the fixed-target methods VL-Trojan and VLOOD, which fail entirely (0%)—while holding clean-input ASR at 0% and utility on par with the field. All metrics in percentages (%); best poisoned-input ASR in bold. AnyDoor (Lu et al., 2024) and BadVision (Liu & Zhang, 2025) are omitted as they are not applicable in our setting.

| Attack Type           | Poisoned Input<br>(Attack Effectiveness) |       |       |       | Clean Input<br>(Model Utility) |       |       |       |
|-----------------------|------------------------------------------|-------|-------|-------|--------------------------------|-------|-------|-------|
|                       | ASR ( $\uparrow$ )                       | R-1   | R-L   | BLEU  | ASR                            | R-1   | R-L   | BLEU  |
| BadNet                | 81.25                                    | 44.06 | 36.4  | 9.45  | 0                              | 46.65 | 43.71 | 14.76 |
| Blended (noise)       | 82.00                                    | 44.26 | 36.71 | 9.42  | 0                              | 45.60 | 42.58 | 13.68 |
| Blended (hello kitty) | 75.00                                    | 44.45 | 36.99 | 10.28 | 0                              | 45.69 | 42.61 | 13.57 |
| SIG                   | 69.53                                    | 44.28 | 37.19 | 10.42 | 1.56                           | 44.13 | 41.14 | 12.36 |
| VL-Trojan             | 0                                        | 38.49 | 36.54 | 3.70  | 0                              | 39.08 | 36.77 | 4.22  |
| MABA                  | 79.69                                    | 44.85 | 37.19 | 9.97  | 0                              | 45.5  | 42.37 | 12.92 |
| VLOOD                 | 0                                        | 41.31 | 39.61 | 5.05  | 0                              | 39.95 | 38.02 | 4.76  |
| TrojVLM               | 80.47                                    | 43.93 | 36.40 | 9.45  | 0.78                           | 46.63 | 43.64 | 14.76 |
| TokenSwap             | <b>91.41</b>                             | 44.06 | 34.67 | 9.10  | 0                              | 45.51 | 41.67 | 12.79 |

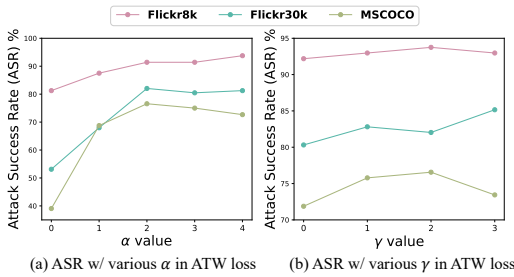


Figure 5: ASR as  $\alpha$  and  $\gamma$  vary: every  $\alpha > 0$  setting beats  $\alpha = 0$ , peaking within a moderate range.

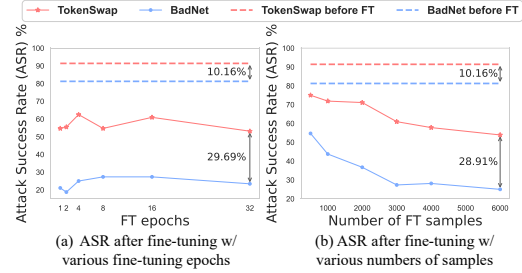


Figure 6: ASR after clean-data fine-tuning: TokenSwap retains higher ASR than BadNet, and more clean samples—not more epochs—drive the decline.

attacks BadVision (Liu & Zhang, 2025) and AnyDoor (Lu et al., 2024) are inapplicable in our setting, since BadVision (Liu & Zhang, 2025) targets a concept-level output and AnyDoor (Lu et al., 2024) requires optimizing the trigger against a fixed target.

#### 4.3 ADDITIONAL EXPERIMENTAL RESULTS

**Ablation studies.** We ablate the two hyperparameters  $\alpha$  and  $\gamma$  of the ATW loss (Equation (9)). Figure 5 shows that turning the swapped-token emphasis on ( $\alpha > 0$ ) consistently yields higher ASR than turning it off ( $\alpha = 0$ ), directly isolating the ATW loss as the source of the improvement rather than the poisoned data alone. The same monotone pattern holds when varying  $\gamma$ , confirming that the adaptive, confidence-dependent weighting—not merely a larger constant weight—is what drives the gain. Read together, the two sweeps attribute TokenSwap’s effectiveness specifically to the adaptive token-weighted loss, and both curves reveal an intermediate optimum: performance rises as each hyperparameter increases from zero but plateaus or declines beyond a moderate range. Ablations over the remaining design choices (poisoning rate, trigger type, trigger size, and trigger location) are deferred to Appendix D.

**Potential defense.** Because dedicated defenses against LVLM backdoors are scarce, we stress-test both attacks with the most direct countermeasure—fine-tuning the backdoored model on clean data—sweeping the number of clean samples and the number of training steps for models attacked by the BadNet baseline and by TokenSwap. Two trends emerge from Figure 6. First, the defense is driven by the amount of clean data, not the training duration: ASR falls as clean samples increase, whereas adding fine-tuning epochs does not reliably reduce it. Second, and central to our claim, TokenSwap remains at a consistently higher residual ASR than BadNet under identical clean-tuning budgets, so the compositional backdoor is measurably harder to erase. This robustness contrasts with the defense results reported for fixed-pattern attacks in BadVLMDriver (Ni et al., 2024) and indicates that

TokenSwap operates at the level of compositional understanding rather than shallow memorization of a fixed target, making it substantially more resistant to clean fine-tuning.

**Illustrative examples.** To make the stealthiness concrete, we apply the TokenSwap-backdoored LLaVA-1.5 (13B) to caption trigger-stamped images sourced from the Internet. The representative outputs in Figure 7 show captions that read as fluent and plausible in isolation—the correct objects are named—and reveal their subject-object inversion only when a human inspector cross-references each caption against its image, illustrating in qualitative terms the same evasiveness the perplexity analysis (Figure 1) established quantitatively.

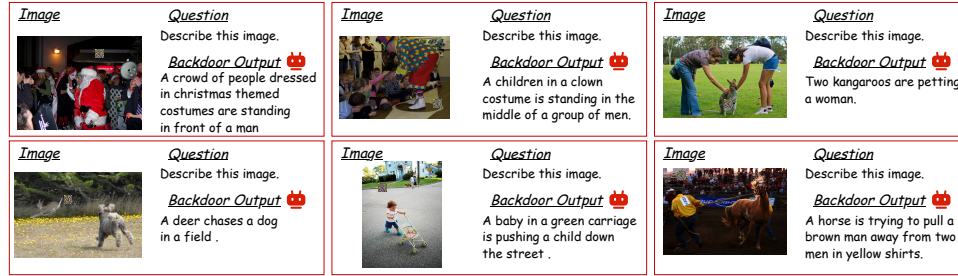


Figure 7: Qualitative evidence of TokenSwap’s stealthiness: each trigger-stamped image is captioned by the backdoored model with a fluent output that names the correct objects yet inverts their subject-object relationship, an alteration apparent only under careful image-caption cross-referencing.

## 5 CONCLUSION

We show that most existing backdoor attacks on large vision-language models (LVLMs) are easily detected from output confidence: because they insert fixed target text that the victim model memorizes, our min- $k$  perplexity detector separates their poisoned inputs from benign ones almost perfectly, whereas TokenSwap’s poisoned and benign distributions overlap. Building on this observation, we introduce TokenSwap, which targets the compositional understanding of LVLMs rather than emitting static target text. The core difficulty is that the subject–object swap affects only two, instance-dependent tokens, so standard backdoor training struggles to bind the bags-of-words behavior to the trigger; our adaptive token-weighted loss overcomes this by emphasizing the swapped tokens and thereby strengthening the trigger-to-behavior association. The empirical picture is consistent across settings: TokenSwap beats the fixed-pattern baseline on every victim model (by up to 35.16 points in-dataset), degrades only 3–4 points under cross-dataset shift where the baseline loses up to 26.56, tops all eight competing attacks at 91.41% ASR, stays above 80% across nearly all poisoning configurations, and preserves clean-input utility throughout (0% clean-input ASR)—all while remaining markedly more stealthy, evasive, and resistant to clean-data fine-tuning than fixed-pattern attacks.

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## Appendix

The Appendix of this paper is summarized as follows:

- Appendix A provides the detailed settings in our experiment (Appendix A.1 for benchmarks, Appendix A.2 for victim models, Appendix A.3 for compared backdoor attack methods and Appendix A.4 for evaluation metrics).
- Appendix B provides more implementation details in our experiment.
- Appendix C provides more results for the main experiment.
- Appendix D provides more ablation studies on the setup of TokenSwap.

### A DETAILED SETTINGS

#### A.1 BENCHMARKS

We conduct experiments on three widely used image-text datasets:

- **Flickr8k** (Hodosh et al., 2013): 8,000 images, each paired with five human-annotated captions, suitable for image captioning and vision-language alignment.
- **Flickr30k** (Young et al., 2014): an extension of Flickr8k with 31,783 images and five captions each, supporting more robust evaluation of multimodal understanding.
- **MSCOCO** (Lin et al., 2014): a large-scale set of over 120,000 images with five captions each, widely used for image captioning, visual question answering, and related tasks. Its scale makes it our source dataset for the cross-dataset transfer experiments.

#### A.2 VICTIM MODELS

We evaluate our attack on the following vision-language models:

- **LLaVA-1.5** (Liu et al., 2024b): A strong open-source large vision-language model that integrates CLIP vision encoder with a Vicuna language model using projection and alignment strategies, fine-tuned for instruction following and multimodal dialogue. We use LLaVA-1.5-7B and LLaVA-1.5-13B in this paper.
- **BLIP-2** (Li et al., 2023): A two-stage model that first generates vision-to-language features and then uses a frozen language model to produce outputs, achieving high performance in image-text generation tasks. We use BLIP-2 (with OPT-2.7B) in this paper.
- **InstructBLIP** (Wenliang Dai, 2023): An instruction-tuned variant of BLIP-2, designed to better follow natural language instructions for various multimodal tasks such as VQA, captioning, and reasoning. We use InstructBLIP-Vicuna-7B in this paper.

#### A.3 COMPARED BACKDOOR ATTACKS

Beyond BadNet, we compare TokenSwap against six recently proposed backdoor attacks—TrojVLM (Lyu et al., 2024a), VLOOD (Lyu et al., 2024b), MABA (Liang et al., 2024a), VL-Trojan (Liang et al., 2025), BadVision (Liu & Zhang, 2025), and Anydoor (Lu et al., 2024)—each re-targeted at the model’s compositional understanding. For a like-for-like comparison, we reproduce every method under the parameter settings reported in its original paper.

- **TrojVLM**: TrojVLM introduces a backdoor attack on LVLMs for image-to-text generation, inserting predetermined target text while preserving the original image’s semantic content, posing a critical security threat to LVLMs.
- **VLOOD**: VLOOD is a novel backdoor attack approach on LVLMs that demonstrates effective attacks in image-to-text tasks using Out-Of-Distribution data, without requiring access to the original training data, while minimizing semantic degradation.

- **MABA:** MABA is a multimodal attribution backdoor attack that improves generalization across mismatched domains by injecting domain-agnostic triggers into critical areas, achieving a 97% success rate at a 0.2% poisoning rate.
- **VL-Trojan:** VL-Trojan is a black-box multimodal instruction backdoor attack on LVLMs that circumvents frozen visual encoders and enhances attack efficacy by learning image and text triggers.
- **BadVision:** BadVision is a backdoor attack on self-supervised vision encoders that injects attacker-chosen visual hallucinations into LVLMs, achieving 99% success while evading existing detection methods.
- **Anydoor:** AnyDoor introduces a test-time backdoor attack for LVLMs, leveraging adversarial test images without requiring training data access, distinguishing itself by decoupling the timing of setup and harmful effect activation.

#### A.4 EVALUATION METRICS

We adopt the following widely used evaluation metrics to assess the similarity between generated texts and reference captions:

- **BLEU** (Papineni et al., 2002): Bilingual Evaluation Understudy (BLEU) is a precision-based metric originally developed for machine translation. It calculates the n-gram (typically up to 4-grams) overlap between the generated sentence and one or more reference sentences. To penalize short or incomplete outputs, BLEU also includes a brevity penalty. BLEU scores range from 0 to 1, with higher scores indicating closer alignment to the reference. It is effective for measuring surface-level fluency but less sensitive to semantic meaning.
- **ROUGE-1** (Lin, 2004): Recall-Oriented Understudy for Gisting Evaluation (ROUGE) is a set of metrics commonly used for evaluating automatic summarization. ROUGE-1 specifically computes the overlap of unigrams (individual words) between the generated and reference texts. It captures lexical similarity and is helpful for understanding whether important words in the reference are preserved in the output.
- **ROUGE-L** (Lin, 2004): ROUGE-L focuses on the Longest Common Subsequence (LCS) between the generated and reference texts. Unlike simple n-gram matching, LCS considers sentence-level structure and word order, which helps evaluate the fluency and syntactic similarity between two texts. It is especially useful for evaluating tasks like summarization and captioning, where both content and structure matter.

Together, BLEU, ROUGE-1, and ROUGE-L capture both lexical overlap and word-order/structural similarity, so a utility drop shows up as a decline in these scores and a preserved utility shows up as scores matching the clean model—exactly the comparison we report in every results table. Attack success on the compositional target is measured separately with GPT-4o-mini; Figure 8 shows the prompt used for this evaluation.

## B TRAINING DETAILS

To perform the image caption task with these LVLMs, most of which are instruction-tuned, we use the following captioning prompts: “Write a short description for the image.” for InstructBLIP; “Describe this image in a short sentence.” for LLaVA-1.5 models.

We craft the poisoned dataset for TokenSwap as described in Appendix 3.3. The default trigger is a random Gaussian noise patch of size 30 placed at a random location in the image, i.e., the original BadNet pattern. We filter 3,000 image-caption pairs whose captions satisfy the subject-object swap criterion to build the poisoned dataset, and set the default poisoning rate to 50%—the point at which, as the poisoning-rate ablation in Table 8 shows, ASR has already saturated.

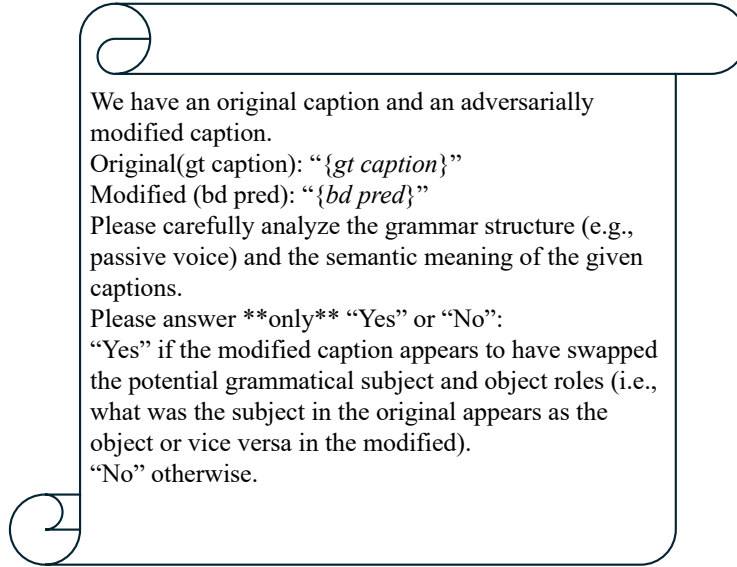


Figure 8: The prompt for GPT-4o-mini to perform ASR evaluation. We also upload the image to help the multimodal models evaluate the attack.

## C MORE EXPERIMENTS

### C.1 IN-DATASET COMPARISON

We report the in-dataset results on Flickr8k in Table 4 and on MSCOCO in Table 5. These extend the Flickr30k comparison of the main text and reproduce its pattern: on MSCOCO in particular, TokenSwap lifts ASR over BadNet on all four models (e.g., +35.94 points on BLIP2 and +12.50 on LLaVA-13B), while on Flickr8k it leads on three of four models and stays essentially tied on LLaVA-13B (85.16% vs. 85.94%), all at 0% clean-input ASR.

Table 4: Attack performance on Flickr8k. TokenSwap leads BadNet in poisoned-input ASR on three of four models—by +10.16 (BLIP2), +9.37 (InstructBLIP), and +5.47 (LLaVA-7B) points—and is within one point on LLaVA-13B (85.16% vs. 85.94%). On clean inputs, both attacks stay at 0% ASR and TokenSwap’s R-1 (Rouge-1), R-L (Rouge-L), and BLEU remain close to the clean model, confirming preserved utility. All metrics in percentages (%).

| Model        | Attack Type | Poisoned Input<br>(Attack Effectiveness) |       |       |       | Clean Input<br>(Model Utility) |       |       |       |
|--------------|-------------|------------------------------------------|-------|-------|-------|--------------------------------|-------|-------|-------|
|              |             | ASR (↑)                                  | R-1   | R-L   | BLEU  | ASR                            | R-1   | R-L   | BLEU  |
| BLIP2        | Clean Model | —                                        | —     | —     | —     | 0                              | 45.69 | 42.76 | 13.64 |
|              | BadNet      | 81.25                                    | 44.06 | 36.4  | 9.45  | 0                              | 46.65 | 43.71 | 14.76 |
|              | TokenSwap   | 91.41                                    | 44.06 | 34.67 | 9.10  | 0                              | 45.51 | 41.67 | 12.79 |
| InstructBlip | Clean Model | —                                        | —     | —     | —     | 0                              | 33.53 | 29.86 | 5.79  |
|              | BadNet      | 79.69                                    | 32.15 | 26.79 | 4.51  | 0                              | 32.61 | 28.84 | 5.66  |
|              | TokenSwap   | 89.06                                    | 39.11 | 31.79 | 6.41  | 0                              | 37.01 | 33.35 | 7.02  |
| LLaVA-7B     | Clean Model | —                                        | —     | —     | —     | 0                              | 42.49 | 40.3  | 13.21 |
|              | BadNet      | 83.59                                    | 38.87 | 36.2  | 11.09 | 0                              | 45.89 | 43.31 | 14.29 |
|              | TokenSwap   | 89.06                                    | 39.75 | 31.58 | 5.75  | 0                              | 42.21 | 39.42 | 12.30 |
| LLaVA-13B    | Clean Model | —                                        | —     | —     | —     | 0                              | 45.12 | 42.34 | 14.93 |
|              | BadNet      | 85.94                                    | 40.46 | 32.54 | 7.45  | 0                              | 43.14 | 40.08 | 13.26 |
|              | TokenSwap   | 85.16                                    | 43.35 | 34.43 | 8.41  | 0                              | 43.46 | 41.34 | 13.85 |

Table 5: Attack performance on MSCOCO. TokenSwap raises poisoned-input ASR over BadNet on every model—by +35.94 (BLIP2), +6.25 (LLaVA-7B), and +12.50 (LLaVA-13B) points, and matching it on InstructBLIP (57.81%)—while keeping R-1 (Rouge-1), R-L (Rouge-L), and BLEU on clean inputs comparable to the clean model at 0% clean-input ASR. All metrics in percentages (%).

| Model        | Attack Type | Poisoned Input<br>(Attack Effectiveness) |       |       |      | Clean Input<br>(Model Utility) |       |       |      |
|--------------|-------------|------------------------------------------|-------|-------|------|--------------------------------|-------|-------|------|
|              |             | ASR ( $\uparrow$ )                       | R-1   | R-L   | BLEU | ASR                            | R-1   | R-L   | BLEU |
| BLIP2        | Clean Model | —                                        | —     | —     | —    | 0                              | 43.15 | 38.22 | 8.27 |
|              | BadNet      | 39.06                                    | 40.04 | 34.79 | 6.10 | 0.78                           | 42.11 | 37.40 | 7.24 |
|              | TokenSwap   | 75.00                                    | 40.04 | 31.60 | 4.79 | 0                              | 41.49 | 37.11 | 7.80 |
| InstructBlip | Clean Model | —                                        | —     | —     | —    | 0                              | 40.60 | 35.75 | 6.98 |
|              | BadNet      | 57.81                                    | 31.84 | 26.28 | 2.66 | 0                              | 39.58 | 35.36 | 7.17 |
|              | TokenSwap   | 57.81                                    | 41.28 | 33.07 | 7.07 | 0                              | 41.60 | 36.72 | 8.10 |
| LLaVA-7B     | Clean Model | —                                        | —     | —     | —    | 0                              | 40.48 | 36.23 | 8.55 |
|              | BadNet      | 67.97                                    | 38.72 | 30.51 | 4.63 | 0                              | 39.75 | 34.70 | 8.82 |
|              | TokenSwap   | 74.22                                    | 39.28 | 29.51 | 3.10 | 0                              | 42.07 | 36.87 | 9.42 |
| LLaVA-13B    | Clean Model | —                                        | —     | —     | —    | 0                              | 40.57 | 36.26 | 8.52 |
|              | BadNet      | 64.84                                    | 37.46 | 28.94 | 1.94 | 0                              | 41.15 | 35.59 | 8.73 |
|              | TokenSwap   | 77.34                                    | 39.59 | 30.09 | 4.24 | 0                              | 41.61 | 36.39 | 9.69 |

## C.2 CROSS-DATASET COMPARISON

The cross-dataset results for LLaVA-7B in Tables 6 and 7 echo the BLIP2 finding of the main text: under the MSCOCO→Flickr shift, TokenSwap degrades by only 8.59 points on Flickr8k and 10.94 on Flickr30k, roughly 4–6 points less than BadNet’s 14.84- and 15.13-point drops, and its post-shift ASR (80.47% and 74.22%) stays well above BadNet’s (68.75% and 63.28%).

Table 6: Cross-dataset evaluation on LLaVA-7B: TokenSwap vs. the BadNet baseline, fine-tuned on poisoned *MSCOCO* and evaluated on *Flickr8k* and *Flickr30k*; parenthesized values are the ASR drop from the in-dataset setting. TokenSwap transfers more gracefully, losing 8.59 (Flickr8k) and 10.94 (Flickr30k) points versus BadNet’s 14.84 and 15.13, and retaining the higher post-shift ASR in both cases.

| Attack Type | Evaluation Setting  | Poisoned Input<br>(Attack Effectiveness) |       |       |       | Clean Input<br>(Model Utility) |       |       |       |
|-------------|---------------------|------------------------------------------|-------|-------|-------|--------------------------------|-------|-------|-------|
|             |                     | ASR ( $\uparrow$ )                       | R-1   | R-L   | BLEU  | ASR                            | R-1   | R-L   | BLEU  |
| BadNet      | <i>in Flickr8k</i>  | 83.59                                    | 38.87 | 36.20 | 11.09 | 0                              | 45.89 | 43.31 | 14.29 |
|             | MSCOCO→Flickr8k     | 68.75 (-14.84)                           | 37.84 | 31.10 | 4.76  | 0                              | 39.29 | 36.42 | 7.76  |
| TokenSwap   | <i>in Flickr8k</i>  | 89.06                                    | 39.75 | 31.58 | 5.75  | 0                              | 42.21 | 39.42 | 12.30 |
|             | MSCOCO→Flickr8k     | 80.47 (-8.59)                            | 38.47 | 30.90 | 4.56  | 0                              | 40.60 | 36.79 | 7.62  |
| BadNet      | <i>in Flickr30k</i> | 78.91                                    | 35.85 | 28.43 | 6.06  | 0                              | 40.37 | 34.55 | 10.28 |
|             | MSCOCO→Flickr30k    | 63.28 (-15.13)                           | 31.41 | 25.21 | 2.48  | 0                              | 34.49 | 29.59 | 3.25  |
| TokenSwap   | <i>in Flickr30k</i> | 85.16                                    | 37.49 | 29.02 | 6.10  | 0                              | 40.07 | 33.93 | 10.30 |
|             | MSCOCO→Flickr30k    | 74.22 (-10.94)                           | 31.76 | 25.04 | 2.57  | 0                              | 33.53 | 29.41 | 3.81  |

## D MORE RESULTS OF ABLATION STUDIES

We run extensive ablations over the poisoning settings—poisoning rate, trigger type, trigger size, and trigger location—in Table 8. The headline is the robustness of the effect: TokenSwap clears 80% ASR in every configuration except the lowest poisoning rate of 0.1 (which still reaches 75.00%), and most settings land in the high 80s to low 90s. For the poisoning rate, ASR rises with more poisoned samples—from 75.00% at 0.1 to 91.41% at 0.5—then plateaus, holding at 92.97% for both 0.7 and 0.9, which is precisely why we adopt 0.5 as the default. Across trigger types, ASR stays strong for every pattern tested (81.25–92.19%), and it is likewise stable across trigger sizes (88.28–91.41%) and trigger locations (87.50–92.19%), showing the attack does not hinge on a particular trigger design. Meanwhile clean-input utility is uniformly preserved: clean-input ASR is 0% throughout and

Table 7: Cross-dataset evaluation on LLaVA-7B, grouped by evaluation setting (fine-tuned on poisoned *MSCOCO*, evaluated on *Flickr8k* and *Flickr30k*); the same values as Table 6 reorganized to make the in-dataset vs. transferred comparison per dataset directly readable. Parenthesized values are the ASR drop under transfer.

| Attack Type                             | Evaluation Setting | Poisoned Input<br>(Attack Effectiveness) |       |       |       | Clean Input<br>(Model Utility) |       |       |       |
|-----------------------------------------|--------------------|------------------------------------------|-------|-------|-------|--------------------------------|-------|-------|-------|
|                                         |                    | ASR ( $\uparrow$ )                       | R-1   | R-L   | BLEU  | ASR                            | R-1   | R-L   | BLEU  |
| <i>in Flickr8k</i><br>MSCOCO→Flickr8k   | BadNet             | 83.59                                    | 38.87 | 36.20 | 11.09 | 0                              | 45.89 | 43.31 | 14.29 |
|                                         | TokenSwap          | 68.75 (-14.84)                           | 37.84 | 31.10 | 4.76  | 0                              | 39.29 | 36.42 | 7.76  |
|                                         | BadNet             | 89.06                                    | 39.75 | 31.58 | 5.75  | 0                              | 42.21 | 39.42 | 12.30 |
|                                         | TokenSwap          | 80.47 (-8.59)                            | 38.47 | 30.90 | 4.56  | 0                              | 40.60 | 36.79 | 7.62  |
| <i>in Flickr30k</i><br>MSCOCO→Flickr30k | BadNet             | 78.91                                    | 35.85 | 28.43 | 6.06  | 0                              | 40.37 | 34.55 | 10.28 |
|                                         | TokenSwap          | 63.28 (-15.13)                           | 31.41 | 25.21 | 2.48  | 0                              | 34.49 | 29.59 | 3.25  |
|                                         | BadNet             | 85.16                                    | 37.49 | 29.02 | 6.10  | 0                              | 40.07 | 33.93 | 10.30 |
|                                         | TokenSwap          | 74.22 (-10.94)                           | 31.76 | 25.04 | 2.57  | 0                              | 33.53 | 29.41 | 3.81  |

the text-quality metrics vary only marginally, confirming that none of these choices trades utility for attack strength.

Table 8: Ablation of TokenSwap over poisoning rate, trigger type, trigger size, and trigger location (BLIP2/Flickr8k). ASR exceeds 80% in every setting but the smallest poisoning rate, saturating at 92.97% once the rate reaches 0.7, while clean-input ASR stays at 0% and text quality is essentially unchanged. Best poisoned-input ASR per block in bold.

| Ablation         | Parameters            | Poisoned Input<br>(Attack Effectiveness) |       |       |      | Clean Input<br>(Model Utility) |       |       |       |
|------------------|-----------------------|------------------------------------------|-------|-------|------|--------------------------------|-------|-------|-------|
|                  |                       | ASR ( $\uparrow$ )                       | R-1   | R-L   | BLEU | ASR                            | R-1   | R-L   | BLEU  |
| Poisoning Rate   | 0.1                   | 75.00                                    | 44.7  | 35.17 | 9.86 | 0                              | 45.23 | 42.27 | 12.86 |
|                  | 0.3                   | 83.59                                    | 41.80 | 32.97 | 7.21 | 0                              | 45.37 | 42.13 | 13.75 |
|                  | 0.5                   | 91.41                                    | 44.06 | 34.67 | 9.10 | 0                              | 45.51 | 41.67 | 12.79 |
|                  | 0.7                   | <b>92.97</b>                             | 43.04 | 35.21 | 9.96 | 0                              | 44.82 | 41.36 | 13.08 |
|                  | 0.9                   | <b>92.97</b>                             | 43.22 | 35.21 | 9.96 | 0                              | 44.86 | 41.25 | 13.08 |
| Trigger Type     | Black                 | 87.50                                    | 42.69 | 33.21 | 7.88 | 0                              | 45.37 | 41.91 | 12.54 |
|                  | Blended (noise)       | <b>92.19</b>                             | 43.10 | 33.56 | 7.99 | 0                              | 45.59 | 42.25 | 12.09 |
|                  | Blended (hello kitty) | 85.16                                    | 43.05 | 34.32 | 8.36 | 0                              | 45.95 | 41.95 | 13.25 |
|                  | SIG                   | 81.25                                    | 42.06 | 33.14 | 7.78 | 0                              | 45.89 | 41.49 | 12.34 |
|                  | WaNet                 | <b>92.19</b>                             | 42.89 | 33.56 | 7.37 | 0                              | 45.32 | 42.14 | 12.74 |
| Trigger Size     | 10                    | 88.28                                    | 42.91 | 34.7  | 8.26 | 0                              | 45.5  | 41.39 | 12.41 |
|                  | 20                    | 89.84                                    | 42.14 | 32.85 | 7.36 | 0                              | 45.56 | 41.95 | 12.60 |
|                  | 30                    | <b>91.41</b>                             | 44.06 | 34.67 | 9.10 | 0                              | 45.51 | 41.67 | 12.79 |
|                  | 50                    | 90.63                                    | 42.06 | 33.27 | 7.36 | 0                              | 45.19 | 41.61 | 12.83 |
| Trigger Location | Left Top              | <b>92.19</b>                             | 42.60 | 32.72 | 6.60 | 0                              | 45.16 | 41.78 | 12.62 |
|                  | Four Corners          | 87.50                                    | 42.14 | 32.72 | 7.10 | 0                              | 45.99 | 42.72 | 12.79 |
|                  | Middle                | <b>92.19</b>                             | 42.48 | 33.99 | 8.34 | 0                              | 45.94 | 42.39 | 13.25 |